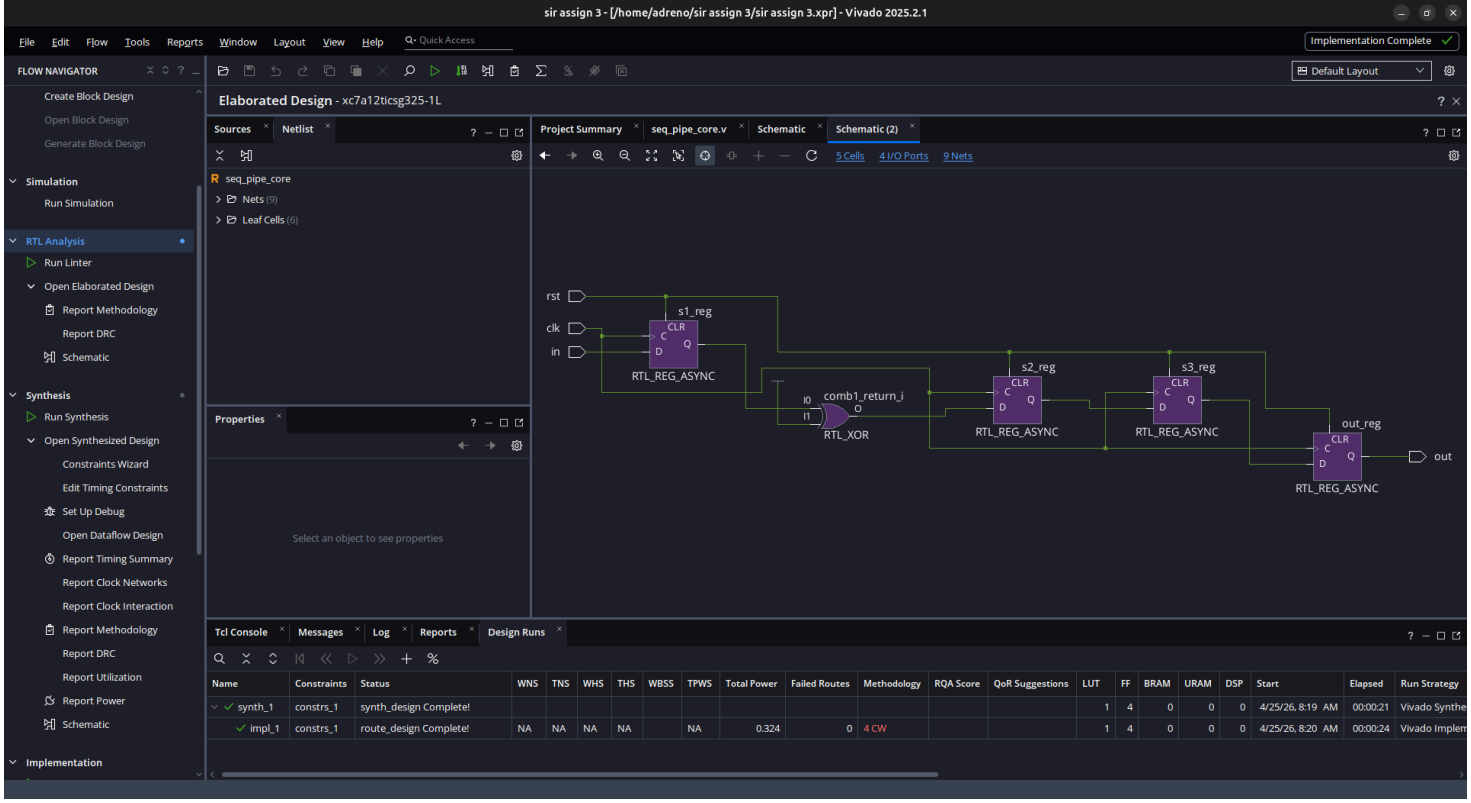
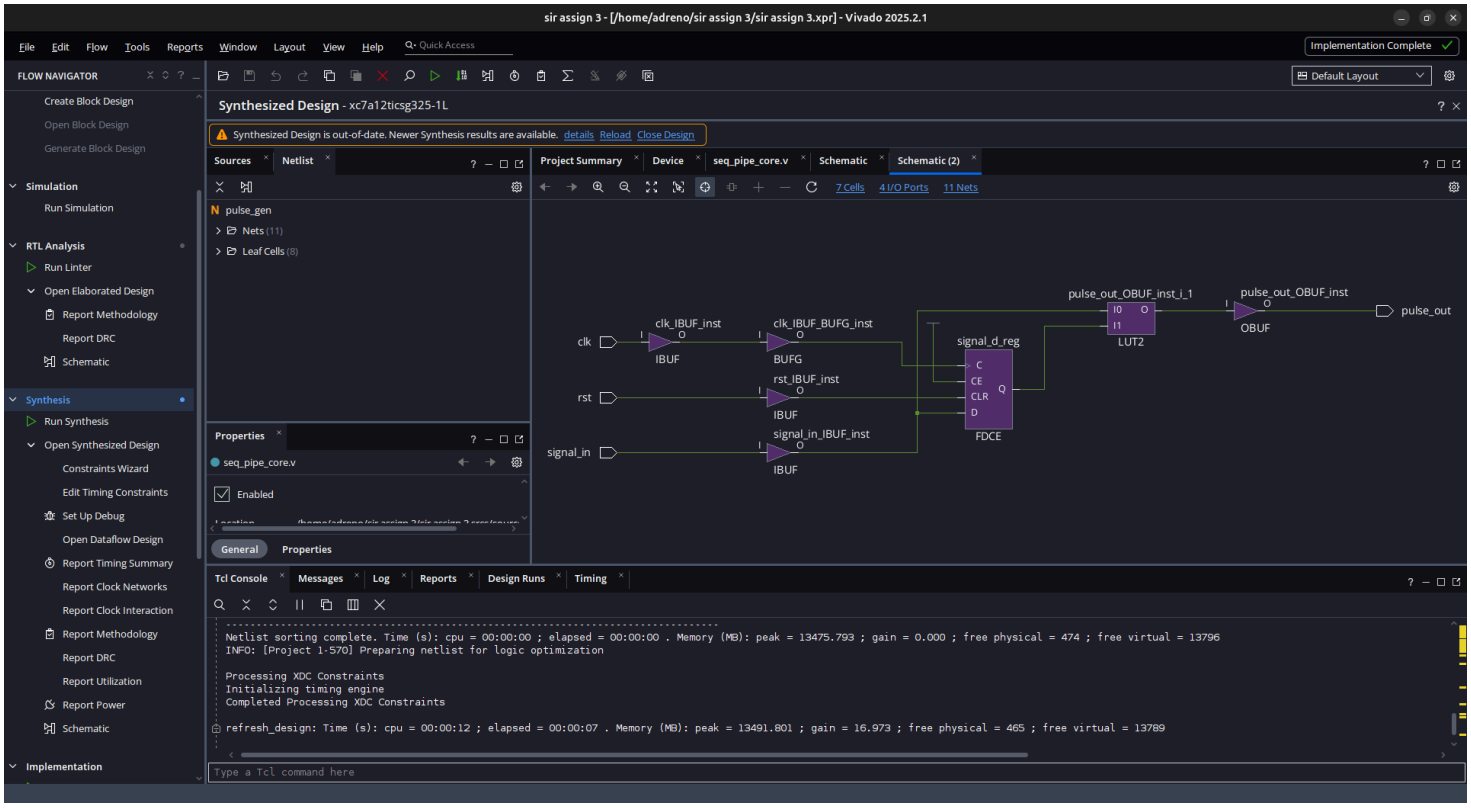
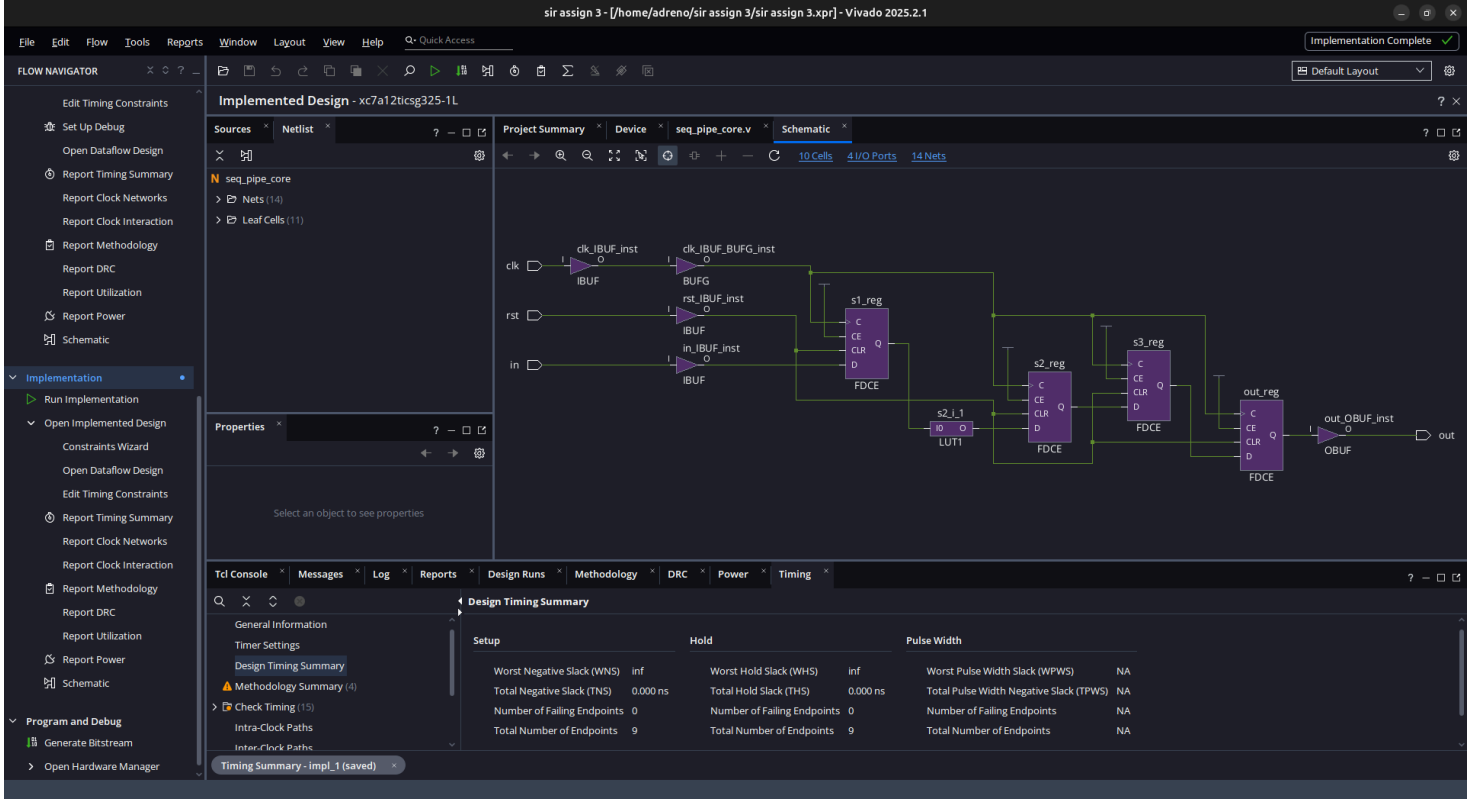
Pipe Line

```
module seq_pipe_core (
input clk,
input rst,
input in,
output reg out
);

reg s1, s2, s3;

always @(posedge clk or posedge rst)
if (rst) s1 <= 0;
else s1 <= in;

always @(posedge clk or posedge rst)
if (rst) s2 <= 0;
else s2 <= comb1(s1);

always @(posedge clk or posedge rst)
if (rst) s3 <= 0;
else s3 <= comb2(s2);

always @(posedge clk or posedge rst)
if (rst) out <= 0;
else out <= s3;

function comb1;
input x;
begin
comb1 = x ^ 1'b1;
end
endfunction

function comb2;
input x;
begin
comb2 = x & 1'b1;
end
endfunction

endmodule
```


sir assign 3 - [/home/adreno/sir assign 3/sir assign 3.xpr] - Vivado 2025.2.1
File Edit Flow Tools Reports Window Layout View Help
Implementation Complete
FLOW NAVIGATOR
Create Block Design
Open Block Design
Generate Block Design
Simulation
Run Simulation
RTL Analysis
Run Linter
Open Elaborated Design
Report Methodology
Report DRC
Schematic
Synthesis
Run Synthesis
Open Synthesized Design
Constraints Wizard
Edit Timing Constraints
Set Up Debug
Open Dataflow Design
Report Timing Summary
Report Clock Networks
Report Clock Interaction
Report Methodology
Report DRC
Report Utilization
Report Power
Schematic
Implementation
Default Layout
Elaborated Design - xc7a12ticsg325-1L
Sources
Netlist
seq_pipe_core
Nets (9)
Leaf Cells (6)
Properties
Select an object to see properties
Project Summary
seq_pipe_core.v
Schematic
Schematic (2)
5 Cells
4 I/O Ports
9 Nets
rst
clk
in
s1_reg
RTL_REG_ASYNC
comb1_return_i
RTL_XOR
s2_reg
RTL_REG_ASYNC
s3_reg
RTL_REG_ASYNC
out_reg
RTL_REG_ASYNC
out
Tcl Console
Messages
Log
Reports
Design Runs
Name | Constraints | Status | WNS | TNS | WHS | THS | WBSS | TPWS | Total Power | Failed Routes | Methodology | RQA Score | QoR Suggestions | LUT | FF | BRAM | URAM | DSP | Start | Elapsed | Run Strategy
synth_1 | constrs_1 | synth_design Complete! | 1 | 4 | 0 | 0 | 0 | 4/25/26, 8:19 AM | 00:00:21 | Vivado Synthe
impl_1 | constrs_1 | route_design Complete! | NA | NA | NA | NA | NA | 0.324 | 0 | 4 CW | 1 | 4 | 0 | 0 | 0 | 4/25/26, 8:20 AM | 00:00:24 | Vivado Implem


sir assign 3 - [/home/adreno/sir assign 3/sir assign 3.xpr] - Vivado 2025.2.1
File Edit Flow Tools Reports Window Layout View Help
Implementation Complete
FLOW NAVIGATOR
Create Block Design
Open Block Design
Generate Block Design
Simulation
Run Simulation
RTL Analysis
Run Linter
Open Elaborated Design
Report Methodology
Report DRC
Schematic
Synthesis
Run Synthesis
Open Synthesized Design
Constraints Wizard
Edit Timing Constraints
Set Up Debug
Open Dataflow Design
Report Timing Summary
Report Clock Networks
Report Clock Interaction
Report Methodology
Report DRC
Report Utilization
Report Power
Schematic
Implementation
Default Layout
Synthesized Design - xc7a12ticsg325-1L
Synthesized Design is out-of-date. Newer Synthesis results are available. details Reload Close Design
Sources
Netlist
pulse_gen
Nets (11)
Leaf Cells (8)
Properties
seq_pipe_core.v
Enabled
General
Properties
Project Summary
Device
seq_pipe_core.v
Schematic
Schematic (2)
7 Cells
4 I/O Ports
11 Nets
clk
clk_IBUF_inst
IBUF
clk_IBUF_BUFG_inst
BUFG
rst
rst_IBUF_inst
IBUF
signal_in
signal_in_IBUF_inst
IBUF
signal_d_reg
FDCE
pulse_out_OBUF_inst_i_1
LUT2
pulse_out_OBUF_inst
OBUF
pulse_out
Tcl Console
Messages
Log
Reports
Design Runs
Timing
Netlist sorting complete. Time (s): cpu = 00:00:00 ; elapsed = 00:00:00 . Memory (MB): peak = 13475.793 ; gain = 0.000 ; free physical = 474 ; free virtual = 13796
INFO: [Project 1-570] Preparing netlist for logic optimization
Processing XDC Constraints
Initializing timing engine
Completed Processing XDC Constraints
refresh_design: Time (s): cpu = 00:00:12 ; elapsed = 00:00:07 . Memory (MB): peak = 13491.801 ; gain = 16.973 ; free physical = 465 ; free virtual = 13789
Type a Tcl command here


sir assign 3 - [/home/adreno/sir assign 3/sir assign 3.xpr] - Vivado 2025.2.1
File Edit Flow Tools Reports Window Layout View Help
Quick Access
Implementation Complete
FLOW NAVIGATOR
Edit Timing Constraints
Set Up Debug
Open Dataflow Design
Report Timing Summary
Report Clock Networks
Report Clock Interaction
Report Methodology
Report DRC
Report Utilization
Report Power
Schematic
Implementation
Run Implementation
Open Implemented Design
Constraints Wizard
Open Dataflow Design
Edit Timing Constraints
Report Timing Summary
Report Clock Networks
Report Clock Interaction
Report Methodology
Report DRC
Report Utilization
Report Power
Schematic
Program and Debug
Generate Bitstream
Open Hardware Manager
Default Layout
Implemented Design - xc7a12ticsg325-1L
Sources
Netlist
seq_pipe_core
Nets (14)
Leaf Cells (11)
Properties
Select an object to see properties
Project Summary
Device
seq_pipe_core.v
Schematic
10 Cells
4 I/O Ports
14 Nets
clk
clk_IBUF_inst
IBUF
clk_IBUF_BUFG_inst
BUFG
rst
rst_IBUF_inst
IBUF
in
in_IBUF_inst
IBUF
s1_reg
FDCE
s2_i_1
LUT1
s2_reg
FDCE
s3_reg
FDCE
out_reg
FDCE
out_OBUF_inst
OBUF
out
Tcl Console
Messages
Log
Reports
Design Runs
Methodology
DRC
Power
Timing
General Information
Timer Settings
Design Timing Summary
Methodology Summary (4)
Check Timing (15)
Intra-Clock Paths
Inter-Clock Paths
Design Timing Summary
Setup
Worst Negative Slack (WNS) inf
Total Negative Slack (TNS) 0.000 ns
Number of Failing Endpoints 0
Total Number of Endpoints 9
Hold
Worst Hold Slack (WHS) inf
Total Hold Slack (THS) 0.000 ns
Number of Failing Endpoints 0
Total Number of Endpoints 9
Pulse Width
Worst Pulse Width Slack (WPWS) NA
Total Pulse Width Negative Slack (TPWS) NA
Number of Failing Endpoints NA
Total Number of Endpoints NA
Timing Summary - impl_1 (saved)